How long can a two-cycle quantum conveyor operate?

Guided theory problem **15.0 points**

How to use this booklet

- Read the **Background** (next pages) first. It explains all the physics you need; you do not have to read any paper.
- There are **18 questions** with **15.0 points** in total. Every question box shows its points. Boxes labelled *modelling bridge* are scored like all other questions. This is a public excerpt of a larger problem, so some labels are skipped.
- Every number, formula and assumption that a question uses is printed in this booklet. The **Toolkit** lists mathematical facts that you may use without proof. The **Glossary** explains each technical word.
- No computer and no many-body diagonalization is needed. A calculator helps in F.1.

AVAILABLE — How to read green frames

A green frame states something you may *use*, with its source: a published paper (**MFP [1]**, **H2 [2]**) or an exact identity (**GIVEN TOOL**). Use it exactly as stated; do not stretch it beyond its stated object and scope.

NOT ESTABLISHED FOR MB1 — How to read red frames

ADDED MODEL: a simplified teaching model; it has an exact solution, but it is not the original pump. **GIVEN BOUND**: an assumption supplied for an exercise. **OPEN**: a result that is still missing for the original pump. Red does not mean wrong.

How to read blue frames

A blue **PICTURE** frame shows the set-up of one or more questions and says in plain words what is being asked. It contains no answers.

Points

Part	Questions	Points
A	A.1 (0.6), A.2 (0.9)	1.5
M	M.1 (1.0)	1.0
B	B.1 (0.5), B.2 (0.9), B.4 (0.8)	2.2
C	C.1 (1.2), C.2 (0.6), C.3 (1.2), C.4 (0.9)	3.9
D	D.1 (0.6), D.2 (1.4)	2.0
E	E.1 (0.5), E.2 (1.0), E.3 (0.9)	2.4
F	F.1 (0.7), F.3 (0.8)	1.5
X	X.1 (0.5)	0.5
		15.0

Conventions

Set $\hbar = 1$. Rightward transport is positive. The ideal motion in this problem is leftward, so the target is $Q_* = -1/2$ particle per period. Charge is counted as a number of particles.

Notation (each symbol has one meaning)

K	the modulation amplitude in $J + K\cos(\cdots)$; from Part B on, $K/J = -3/2$
$\mathcal{K}$ (calligraphic)	a generator of time evolution in a chosen frame, e.g. $\mathcal{K}_W$, $\mathcal{K}_{\mathrm{ret}}$
J; $J_j(t)$, $t_s(\phi)$; J_z	the hopping unit; the time-dependent hoppings; the coupling of the A.2 spin chain
I, I', I_X, I_0	a current operator; the same current in a moving frame; a local approximation to it; a bound on its norm
$\mathbb{1}$	the identity operator ($\mathbb{1}_2$: the 2×2 identity)
p	a projector in Part B; a probability in Parts D and E
Y; V, Λ	a 5×2 matrix of two eigenvectors; 2×2 unitary matrices
$\mathcal{F}$	a fidelity (E.1)
T, ϕ, Ω	the drive period; the drive phase $\phi = 2\pi t/T = \Omega t$; $\Omega = 2\pi/T$
R	a cell index, or a separation between cells

Background. Everything you need to know

1. Particles carried by a slow, periodic drive

In a *Thouless pump* the parameters of a lattice Hamiltonian are changed slowly and periodically in time. After each period T the Hamiltonian is the same as before, yet particles can be carried along the lattice. For non-interacting fermions that completely fill a band which is separated by gaps from the other bands, the number of particles carried past any bond in one period is an *integer*. This integer is a topological property of the band (its Chern number), so small changes of the model do not change it.

2. The model of this problem

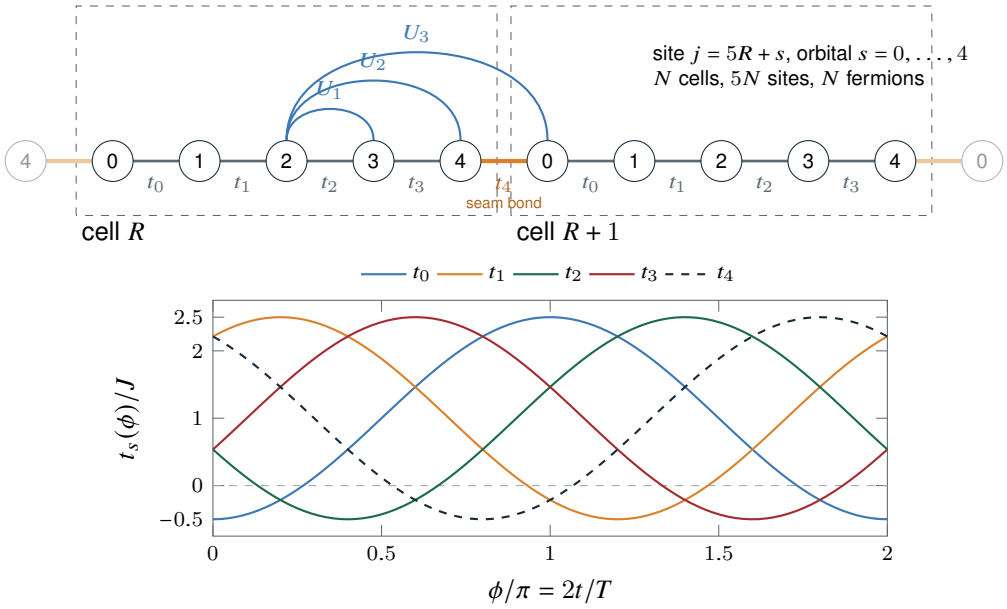
Spinless fermions hop on a chain with five sites per unit cell (Picture 1). There are N cells, $5N$ sites, and N particles: one particle per cell on average ("unit filling"). The Hamiltonian is

$$H_{\mathrm{MFP}}(t) = -\sum_j \left[J_j(t) c_{j+1}^\dagger c_j + \mathrm{h.c.}\right] + \sum_j \sum_{r=1}^{3} U_r n_j n_{j+r},$$

$$J_j(t) = J + K\cos\left(\frac{2\pi t}{T} + \frac{4\pi j}{5}\right), \qquad n_j = c_j^\dagger c_j.$$

Here c_j removes a fermion from site j, and $U_r > 0$ is the repulsion between two particles r sites apart. "MFP" (multi-band fractional pump) is the name of this model from Ref. [1].

PICTURE 1 — The physical system: a driven chain with five sites per cell



Top. Site $j = 5R + s$ is orbital s of cell R. The bond from orbital s to $s+1$ has hopping $t_s(\phi)$. The orange bond t_4 joins orbital 4 of cell R to orbital 0 of cell $R+1$; in M.1 this "seam" bond is the one that picks up the factor $e^{\mp ik}$. Blue arcs: the repulsion U_r between two particles $r = 1, 2, 3$ sites apart. **Bottom.** The drive at $K/J = -3/2$: the five hoppings as the phase $\phi = 2\pi t/T$ advances. After one period every hopping is back to its starting value. The whole problem asks whether particles have nevertheless moved, and for how long that transport survives.

3. Bands

Without interactions, translation by one cell is a symmetry. The one-particle problem then reduces to a 5×5 matrix $h(k, \phi)$ for each quasi-momentum k (you derive it in M.1). Its five eigenvalues, as functions of k, are the five *bands* (Picture 3). At $K/J = -3/2$ the two lowest bands lie close to each other and far below the other three.

4. Localized orbitals and their centres

From all the Bloch states of one band, or of a group of bands, one can build orbitals that are localized in space, one per cell for each band. They are called *Wannier orbitals*. The expectation value $\langle w|x|w\rangle$ of the position in such an orbital is its *centre*. If every orbital of a set is occupied and every centre moves by Δx, the transported charge, counted in particles per cell, is $\Delta x/a$, where a is the cell length. Picture 2(a): orbitals built from band 1 alone move by $+2$ cells per period. This is an ordinary, integer Thouless pump.

5. Two bands together: flavors and a two-cycle conveyor

Mixing bands 1 and 2 gives two localized orbitals per cell, called *flavor* A and *flavor* B. They can be more compact than orbitals of band 1 alone. Picture 2(b) shows the essential geometry: during one period an A orbital slides to the place where a B orbital was. Only after two periods is it back on an A position, one cell to the left. Suppose the particles occupy one flavor only and keep following it. Then every particle moves one cell to the left in two periods: the average transport is $-1/2$ particle per period. This is a *fractional* pump at integer filling.

6. Why interactions matter

Without interactions the N particles fill band 1, and the pump carries $+2$ particles per period. A strong finite-range repulsion prefers compact, well-separated charges. Occupying one flavor in every cell (all A, or all B) is then favorable. These two polarized states exchange their roles every period. Exact numerical solutions of small systems [1] show transport of $-1/2$ per period over a range of interaction strengths and drive frequencies. The drive must be slow enough not to excite other states, but fast enough to follow one flavor rather than to tunnel between the all-A and all-B states.

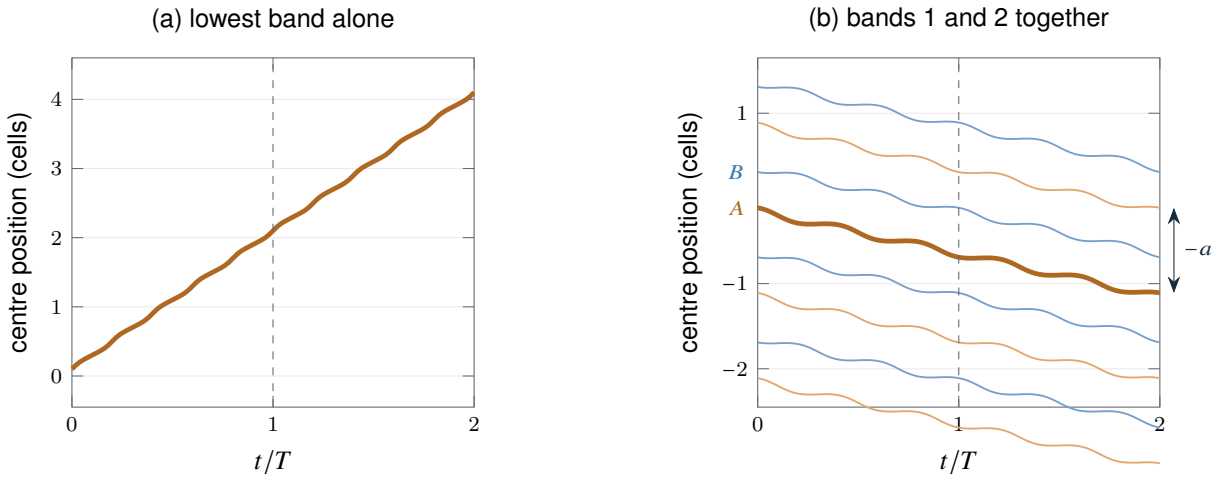
7. The open question

In a long chain, a region of the other flavor (a *domain*) can appear during the drive. Domains destroy the fractional transport. Reference [1] expects the fractional pump of a large system to survive only up to some finite time, and leaves that time open. This problem builds and tests the tools one would need to predict that time:

- **Geometry (Parts M, B).** Describe the moving plane of bands 1 and 2 without choosing a basis inside it (a projector p_{12}), and how it moves (a generator g_t).
- **Frames (Part C).** Describe the many-body state in a frame that follows this geometry (W) and is dressed by correlations (e^S). Keep part of the new generator ($\mathcal{K}_{\mathrm{ret}}$) and bound the rest ($\mathcal{R}$).
- **Current (Part D).** The transported charge is the time integral of the physical current. After a change of frame, the current operator changes too.
- **Local errors (Part E).** In a long chain, compare states only in a region X near the place where the current is measured.
- **Time (Part F).** Turn error bounds into a statement of the form “the transported charge stays within the tolerance until time t”.

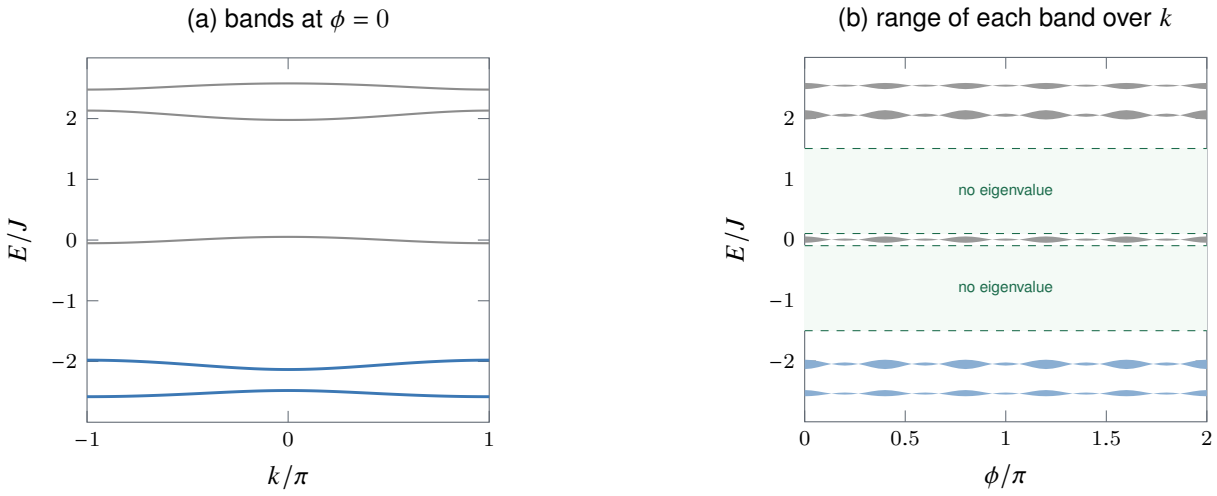
Picture 4 on the reading-map page shows how these steps connect.

PICTURE 2 — Wannier centres: an integer pump and a two-cycle conveyor



Computed for this booklet from $h(k, \phi)$ at $K/J = -3/2$: centres of the localized (Wannier) orbitals, obtained from the projected position operator on a 24-cell ring. This is single-particle geometry only; no interacting dynamics is computed. (a) One orbital per cell built from band 1 alone: it moves by an integer number of cells per period (an ordinary Thouless pump). (b) Two orbitals per cell built from bands 1 and 2 together: flavor A (orange) and flavor B (blue). Follow the thick curve. It does not return to an A position after one period; it returns only after two periods, one cell to the left ($-a$). **A.1 asks:** if every particle rides a curve like the thick one, how much charge has passed after $2T$, and per period on average? Whether interacting particles really stay on one such curve is the open question of the whole problem.

PICTURE 3 — M.1 — the five single-particle bands at $K/J = -3/2$



Computed for this booklet from $h(k, \phi)$. Blue: the two lowest bands, treated together as one group; $p_{12}(k, \phi)$ projects onto this blue pair. Grey: bands 3–5. **M.1 asks** you to derive the 5×5 matrix whose eigenvalues are these curves, and to build p_{12} from the two blue eigenvectors. No eigenvalue enters the two green windows: the blue group stays well separated from band 3 (numerically by about $1.9J$).

Toolkit. Facts you may use without proof

T1. Projectors. An orthogonal projector obeys $p^2 = p = p^\dagger$. Its rank is $\mathrm{Tr}\, p$.

T2. Change of frame. If $i\partial_t|\psi\rangle = H|\psi\rangle$ and $|\xi\rangle = U^\dagger|\psi\rangle$ with a differentiable unitary $U(t)$, then $i\partial_t|\xi\rangle = (U^\dagger H U - iU^\dagger \dot{U})|\xi\rangle$.

T3. Time-dependent operators. If $i\partial_t|\xi\rangle = \mathcal{K}|\xi\rangle$ and $O(t)$ is an operator, then $\frac{\mathrm{d}}{\mathrm{d}t}\langle\xi|O|\xi\rangle = \langle\xi|\big(\partial_t O + i[\mathcal{K}, O]\big)|\xi\rangle$.

T4. Pauli matrices. $\sigma_a\sigma_b = \delta_{ab}\mathbb{1} + i\epsilon_{abc}\sigma_c$; for example $[\sigma_z, \sigma_x] = 2i\sigma_y$. For a unit vector $\boldsymbol{n}$: $e^{ix\boldsymbol{n}\cdot\boldsymbol{\sigma}} = \mathbb{1}\cos x + i(\boldsymbol{n}\cdot\boldsymbol{\sigma})\sin x$. Operators on different spins commute.

T5. Norms. $\|A\|$ is the operator norm (largest singular value); $\|A\|_1$ is the trace norm (sum of singular values). Then $\|AB\| \le \|A\|\,\|B\|$, $\|[A,B]\| \le 2\,\|A\|\,\|B\|$, and $|\,\mathrm{Tr}(AB)| \le \|A\|_1\,\|B\|$. For density matrices $\|\rho - \sigma\|_1 \le 2$.

T6. Reduced states. Split a system into a region X and the rest $\bar{X}$. The reduced state is $\rho_X = \mathrm{Tr}_{\bar{X}}\,\rho$ (trace over $\bar{X}$ only). If O_X acts only on X, then $\mathrm{Tr}(\rho O_X) = \mathrm{Tr}_X(\rho_X O_X)$ and $\mathrm{Tr}_{\bar{X}}(O_X\rho) = O_X\rho_X$.

T7. Fermion modes. $\{c_x, c_y^\dagger\} = \delta_{xy}$. For a one-particle vector f, $a(f) = \sum_x f_x^* c_x$ removes a particle from orbital f, and $\|a(f)\| = \|f\|$. A one-particle matrix A becomes the many-body operator $\widehat{A} = \sum_{xy} A_{xy} c_x^\dagger c_y$. A one-particle unitary U becomes a many-body unitary $\Gamma(U)$ that maps each $c_x^\dagger$ to $\sum_y U_{yx} c_y^\dagger$; it conserves particle number.

T8. Counting. N fermions in M distinct modes can be placed in $\binom{M}{N}$ ways.

T9. Resolvent. For a Hermitian matrix h, $\big\|(\zeta - h)^{-1}\big\| = 1/\mathrm{dist}(\zeta, \text{spectrum of } h)$. If a closed contour $\mathscr{C}$ encloses some eigenvalues and no others, $\frac{1}{2\pi i}\oint_{\mathscr{C}}(\zeta - h)^{-1}\mathrm{d}\zeta$ is the projector onto their eigenvectors.

T10. Fourier coefficients. For a 2π-periodic matrix function $f(k)$, $f_R = \int_{-\pi}^{\pi}\frac{\mathrm{d}k}{2\pi}e^{ikR}f(k)$ and $\|f_R\| \le \int\frac{\mathrm{d}k}{2\pi}\,\|f(k)\|$. If f extends analytically to the strip $|\,\mathrm{Im}\,k| \le \kappa$, the integration line may be moved to $\mathrm{Im}\,k = \pm\kappa$.

T11. Wannier centres in one dimension. On a ring of length L, let P project onto the occupied one-particle states of a group of bands. For a long ring, the eigenvalues of $Pe^{2\pi i x/L}P$ on the range of P are close to $e^{2\pi i x_\alpha/L}$, where x_α are the centres of the maximally localized orbitals of that group. In momentum space this operator is the *Wilson loop*: the ordered product of the overlaps of the band eigenvectors around the Brillouin zone, at fixed drive phase ϕ.

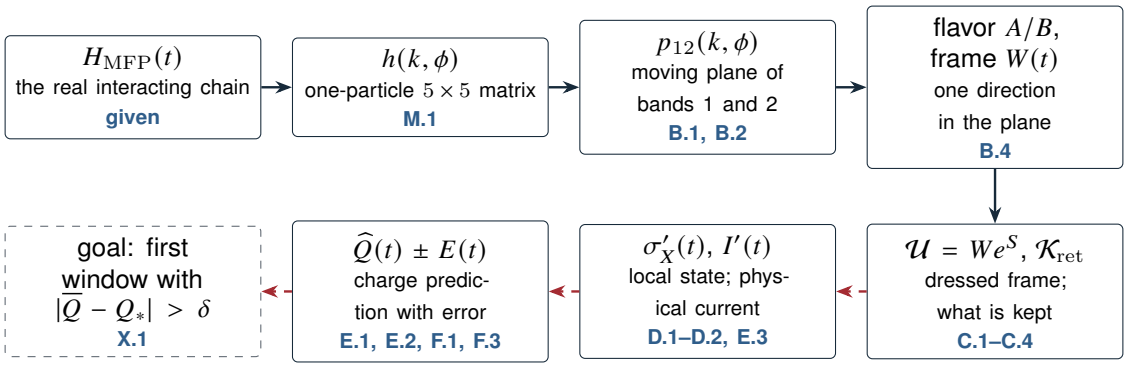
Glossary

Band; Bloch matrix	The eigenvalues $E_n(k,\phi)$ of the one-particle matrix $h(k,\phi)$; h itself (M.1).
Composite bands	Bands 1 and 2 treated as one group, without separating them.
Projector p_{12}	The projector onto the plane spanned by the band-1 and band-2 eigenvectors at given (k,ϕ). $\mathrm{Gr}(2,5)$ is the set of all planes (2-dimensional subspaces) in $\mathbb{C}^5$.
Wannier orbital, centre	A localized orbital built from Bloch states; its mean position.
Flavor A, B	The two localized orbitals per cell built from bands 1 and 2.
Seam	(i) The bond t_4 between neighbouring cells. (ii) In time: the matching at $t = T$, where the labels A and B exchange.
Reference state	A simple state that the true state is compared with, e.g. all particles in flavor A.
Moving frame $W(t)$	A unitary that follows the geometry; the state is described relative to it (T2).
Dressing e^S	A further unitary, with S anti-Hermitian, that builds correlations into the reference.
$\mathcal{K}_{\mathrm{ret}}$, $\mathcal{R}$	The part of the transformed generator that is kept, and the part that is omitted.
Slater determinant; single/double excitation	A state of independent fermions in chosen orbitals; the same state with one or two orbitals replaced by others.
External weight	The probability that a state lies outside a chosen reference space.
Reduced state; partial trace	See T6.
Trace distance	$\|\rho - \sigma\|_1$: how well two states can be told apart by any measurement (at most 2).
Fidelity	$\mathcal{F}(\rho, \|a\rangle) = \langle a\|\rho\|a\rangle$ (E.1).
Window charge $\overline{Q}(t)$	Charge per period, averaged over the two periods that end at t (Part F).
MB1	The research project behind this problem: the original MFP pump at $K/J = -3/2$.

Reading map. What is the object at each step?

Research target: predict when the original interacting pump leaves a chosen charge tolerance, and identify the process that controls that change. A guaranteed interval is useful, but is not automatically the actual first-failure time.

PICTURE 4 — The chain of objects used in this problem



Read the boxes in the order of the arrows. Each box is a mathematical object; the blue label says which questions work with it. Black arrows are steps that this booklet derives. Red dashed arrows are links that are *not yet established* for the original pump; several questions ask why. The interactions of H_{MFP} are not lost after the first arrow: they come back in Part C.

Step	Why the next object is needed	Questions
Target	Ideal winding must be distinguished from actual following.	A.1–A.2
Model object	Five Bloch components are not the full many-body state. Select the composite bands for a physical reason.	M.1
Geometry	Compare a moving subspace without choosing arbitrary basis vectors. Then resolve its internal flavor.	B.1, B.2, B.4
Dynamics	A change of frame does not by itself remove interactions. Decide what to keep and what to bound.	C.1–C.4
Current	Use the transformed physical current, not a population count.	D.1–D.2
Local state	The kept Hamiltonian must actually determine local occupations and coherences.	E.1–E.3
Time	Convert local-state and current errors into a charge interval.	F.1, F.3, X.1

NOT ESTABLISHED FOR MB1 — OPEN — the central dynamical link

The reference evolution and its local error are not yet controlled over the full drive. In particular, $\mathcal{K}_{\mathrm{ret}} \to \sigma'_X \to \widehat{Q}$ is not supplied by the composite-band geometry.

Source / scope: Recommended route: A → M → B → C → D → E → F → X.

Part A. Fractional transport and domain energy (1.5 points)

AVAILABLE — MFP [1] — a candidate two-cycle branch

The two Wannier flavors may exchange after one period. In the source argument, a continuously followed flavor returns after two periods with a one-cell displacement (Background §5, Picture 2). The argument assumes that the occupations continue to follow that branch.

NOT ESTABLISHED FOR MB1 — ADDED IDEAL-FOLLOWING CONDITION

For A.1, stipulate that the actual occupied orbitals follow the branch exactly. This is not a large-system finite-speed stability result.
An N-cell ring has cell length a and contains N particles. After $2T$, each occupied orbital has an unwrapped center displacement $\Delta x_j = -a$. Define the spatially averaged transported particle number by

$$Q(0, 2T) = \frac{1}{Na} \sum_{j=1}^{N} \Delta x_j .$$

The single-period values $Q(0, T)$ and $Q(T, 2T)$ are defined in the same way from the displacements during each period.

A.1 Find $Q(0, 2T)$ and the transport per drive period, averaged over these two periods. Must the two single-period values each equal $-1/2$? Explain. **0.6 pt**

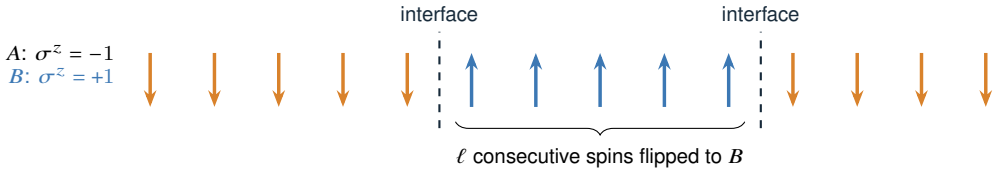
NOT ESTABLISHED FOR MB1 — ADDED MODEL — a spin-chain analogy

For A.2 only, replace the pump by the model below. Its parameters are not identified with K or U_r. This model explains an energy competition; it is not a controlled reduction of MFP.

$$H_{\mathrm{d}} = -J_z \sum_j \sigma_j^z \sigma_{j+1}^z - h \sum_j \sigma_j^z - g \sum_j \sigma_j^x, \qquad J_z, h > 0.$$

Let A mean $\sigma^z = -1$ and B mean $\sigma^z = +1$. Set $g = 0$. Start with all spins in A. Flip a consecutive segment of ℓ spins to B, with $1 \le \ell < N$. There are two interfaces between the segment and its surroundings.

PICTURE 5 — A.2 — a domain in the teaching spin chain



Orange arrows: A ($\sigma^z = -1$). Blue arrows: B ($\sigma^z = +1$). **A.2 asks:** (i) the energy bookkeeping — how much each of the two interfaces costs through the J_z term and how much each flipped spin changes through the h term; (ii) the shortest segment that lowers the energy; (iii) whether lower energy alone forces an isolated chain that starts all-A to produce such a segment when $g = 0$.

A.2 Find the energy change $\Delta E(\ell)$. Find the smallest integer ℓ for which it is strictly negative, provided that segment fits in the chain. Does an isolated system initially in the all-A state then have to form this lower-energy domain when $g = 0$? **0.9 pt**

Part M. From the lattice to the geometric object (1.0 points)

AVAILABLE — MFP [1], Eq. (1)–(2) — source model and motivation

In the source mechanism, mixing the lowest two bands permits more localized Wannier orbitals, and finite-range repulsion can favor occupation of one flavor (Background §5–6).

M.1 input: fix the representation before choosing a projector

Write the physical site index as $j = 5R + s$, with cell R and orbital $s = 0, \ldots, 4$. Define

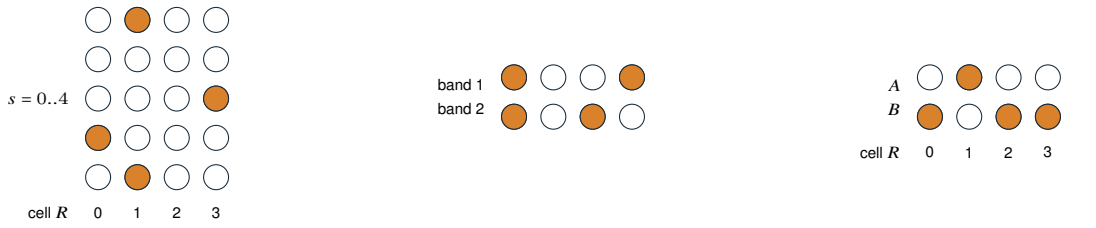
$$t_s(\phi) = J + K\cos(\phi + 4\pi s/5), \qquad c_{R,s} = \frac{1}{\sqrt{N}} \sum_k e^{ikR} c_{k,s},$$

where $k = 2\pi m/N$ is a dimensionless cell momentum. The single-particle part is

$$H_0 = -\sum_R \left[\sum_{s=0}^{3} t_s c^\dagger_{R,s+1} c_{R,s} + t_4 c^\dagger_{R+1,0} c_{R,4} + \text{h.c.} \right].$$

The full model remains $H_{\text{MFP}} = H_0 + H_{\text{int}}$.

PICTURE 6 — M.1(c) — three different ways to place $N = 4$ particles in $N = 4$ cells



(i) any 4 of the $5N$ site orbitals

(ii) any 4 of the $2N$ two-band modes

(iii) exactly one of A, B in every cell

Filled circles: occupied single-particle states (one example pattern each). **M.1(c) asks** you to count all allowed patterns in each picture, and to say which physical restriction each picture imposes. Only after that can you judge whether Fourier transformation has reduced the many-body problem.

M.1 *Modelling bridge* **1.0 pt**

(a) Derive the 5×5 matrix $h(k,\phi)$ in the ordered basis $(0,1,2,3,4)$. Check both complex seam entries and Hermiticity for real k.
(b) Let $Y = (|u_1\rangle, |u_2\rangle)$ contain two orthonormal lowest-band eigenvectors. Construct $p_{12} = YY^\dagger$. Find its rank and ambient dimension. Why is this object physically motivated in MFP?
(c) Compare $\binom{5N}{N}$, $\binom{2N}{N}$, and 2^N. State the different restrictions behind them. Evaluate all three at $N = 4$. Has Fourier transformation solved the many-body problem?

Hint. Insert the Fourier expansion into each term of H_0 and use $\frac{1}{N}\sum_R e^{i(k'-k)R} = \delta_{kk'}$. Write $H_0 = \sum_k \boldsymbol{c}_k^\dagger h(k,\phi)\boldsymbol{c}_k$ with $\boldsymbol{c}_k = (c_{k,0}, \ldots, c_{k,4})^T$.

Useful counting rule

Choosing N occupied modes among M distinct fermionic modes gives $\binom{M}{N}$ occupation states. Exactly one occupied flavor per cell is an additional restriction, not the same as average filling one.

NOT ESTABLISHED FOR MB1 — OPEN — a reference choice is not a dynamical truncation

The two-band N-particle sector is not a two-state system. Restricting the full interacting model to this sector requires control of the terms that leave it.

Connection to the next step: The physical output is a defined composite-band projector. Part B describes its motion. B.4 then asks how to identify a flavor inside it.

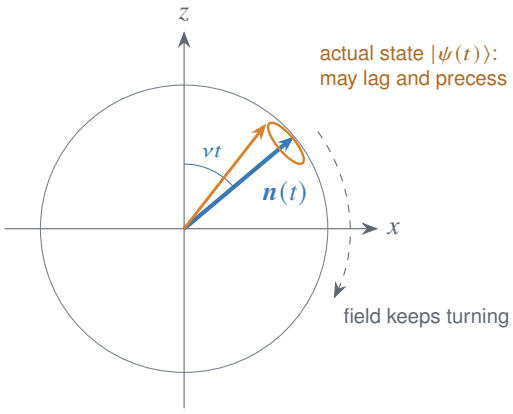
Part B. Geometry and actual motion (2.2 points)

AVAILABLE — H2 [2] — projector kinematics

A projector represents a subspace without choosing a basis inside it. Its derivative describes the change of that subspace. In B.1, derive the corresponding horizontal generator. This is a kinematic statement, not a lifetime prediction.
Let $p(t)$ be a smooth orthogonal projector, so $p^2 = p = p^\dagger$.

PICTURE 7 — B.1 and B.2 — a moving projector and one spin in a turning field

$\boldsymbol{n}(t) = (\sin \nu t, 0, \cos \nu t)$
ground-state projector $p(t) = \frac{1}{2}(\mathbb{1} + \boldsymbol{n} \cdot \boldsymbol{\sigma})$
frame $W(t)$ turns the axes with $\boldsymbol{n}(t)$



The circle is a cut through the Bloch sphere. The field direction $\boldsymbol{n}(t)$ (blue) turns at rate ν; its size Δ is fixed. The instantaneous ground state points along $\boldsymbol{n}$, which is the projector path $p(t)$. **B.1 asks** for a generator g_t that is built from the path $p(t)$ alone and moves it correctly. **B.2 asks** you to go to the frame that turns with $\boldsymbol{n}(t)$, find the Hamiltonian there, and from it the probability that the orange state is found in the excited state. The path depends only on ν; the question is what Δ does.

B.1 Show that $p\dot{p}\,p = 0$. Define $g_t = i[\dot{p}, p]$ and prove $[g_t, p] = i\dot{p}$. State what this identity determines and what it does not determine about the actual Hamiltonian. **0.5 pt**

NOT ESTABLISHED FOR MB1 — ADDED MODEL — an exactly solvable rotating spin

For B.2, study one spin. This is a test of moving-frame dynamics, not the complete interacting pump.

$$H(t) = -\frac{\Delta}{2}\big[\sin(\nu t)\sigma_x + \cos(\nu t)\sigma_z\big], \qquad \Delta, \nu > 0.$$

Its instantaneous ground-state projector and a rotating frame are

$$p(t) = \tfrac{1}{2}\big[\mathbb{1} + \sin(\nu t)\sigma_x + \cos(\nu t)\sigma_z\big],$$
$$W(t) = e^{-i\nu t\sigma_y/2}, \qquad |\Xi(t)\rangle = W^\dagger(t)|\psi(t)\rangle.$$

Use $|\psi(0)\rangle = |\uparrow\rangle$, where $\sigma_z|\uparrow\rangle = |\uparrow\rangle$.

B.2 Find the time-independent moving-frame Hamiltonian $\mathcal{K} = W^\dagger H W - iW^\dagger \dot{W}$. Find the exact probability $P_{\mathrm{ex}}(t)$ of occupying the instantaneous excited state. Explain how changing Δ can change this probability without changing the projector path. **0.9 pt**

Useful identity

For a unit vector $\boldsymbol{n}$ and real x,

$$e^{ix\boldsymbol{n}\cdot\boldsymbol{\sigma}} = \mathbb{1}\cos x + i(\boldsymbol{n}\cdot\boldsymbol{\sigma})\sin x.$$

Source / scope: B.1 uses H2. B.2 is a teaching construction. Its exact solution does not prove a regime of the original pump.

B.4. A plane is not yet a flavor

AVAILABLE — MFP [1], Eq. (2); H2 [2] — two linked objects

The source chooses localized Wannier flavors inside the composite bands. H2 describes the composite projector without an arbitrary internal basis. A physical flavor requires a localization rule and continuous tracking, not just the endpoint statement $p_{12}(T) = p_{12}(0)$.

Let $Y(k,\phi) = (|u_1\rangle, |u_2\rangle)$ and let $V(k,\phi)$ be a unitary 2×2 matrix. If v_α is its column, define

$$u_\alpha = Yv_\alpha, \qquad p_\alpha = Yv_\alpha v_\alpha^\dagger Y^\dagger, \qquad \alpha = A, B.$$

The source's Wannier construction has the form

$$|w_{R,\alpha}\rangle = \frac{1}{\sqrt{N}} \sum_{k,n=1,2} e^{-ikR} V_{n\alpha}(k,\phi)|\psi_{kn}(\phi)\rangle.$$

Here $|\psi_{kn}\rangle$ is a full Bloch state. Its physical center uses the actual site positions $x_{R,s} = Ra + r_s$. The Wilson loop is described in T11.

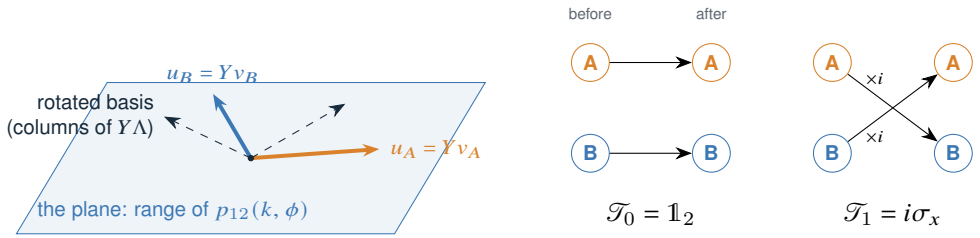
B.4 *Modelling bridge* **0.8 pt**

(a) Show that replacing Y by $Y\Lambda$ for any unitary Λ leaves p_{12} unchanged. Show that a fixed physical p_A also stays unchanged if $v_A \to \Lambda^\dagger v_A$. What changes if one selects a different internal direction without this compensation?
(b) Explain what must be evaluated to obtain the centers $\langle w_{R,\alpha}|x|w_{R,\alpha}\rangle$ and a continuously labelled A/B path. Distinguish the Wilson loop around k from evolution over physical time.
(c) Use the added counterexample below. Does a closed composite projector or a determinant phase determine internal exchange? Does internal exchange alone prove charge transport?

NOT ESTABLISHED FOR MB1 — ADDED COUNTEREXAMPLE — not computed MFP matrices

In a physically labelled two-dimensional space, compare $\mathcal{T}_0 = \mathbb{1}_2$ with $\mathcal{T}_1 = i\sigma_x$. Both act inside the same plane. No real-space displacement is specified. These are illustrative internal maps, not the original pump's Wilson or evolution matrices.

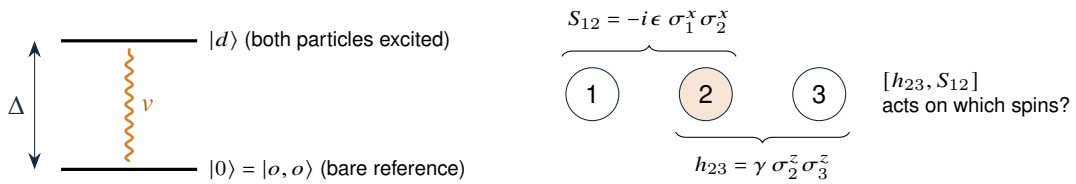
PICTURE 8 — B.4 — a plane is not yet a flavor



Left. The two lowest bands span a plane at each (k, ϕ). Inside it one can pick two orthonormal directions u_A, u_B (the flavors), or any rotated pair. **B.4(a) asks** which choices change only the coordinates and which change a physical object. **B.4(b) asks** what extra information — a localization rule, the real site positions, continuous tracking in time — turns the plane into labelled A/B orbitals with centres. **Right.** Two internal maps of the same plane. **B.4(c) asks** whether one number such as a determinant phase can tell $\mathcal{T}_0$ from $\mathcal{T}_1$, and whether swapping A and B by itself moves charge.

Part C. Correlation and overlapping local corrections (3.9 points)

PICTURE 9 — C.1 and C.2 — one correlation channel; two overlapping corrections



Left (C.1). The bare state $|0\rangle$ couples with strength v to one excited state $|d\rangle$ that lies Δ higher. C.1 asks for the true lowest state of this 2×2 problem, and for the probability $P_d(t)$ of $|d\rangle$ if one starts in $|0\rangle$. Does the short-time growth of P_d define a decay rate? **Right (C.2).** A correction S_{12} designed for spins 1–2 and a Hamiltonian term h_{23} on spins 2–3 share spin 2 (orange). C.2 asks for their commutator and why it matters when corrections designed for separate pairs are put together along a chain.

NOT ESTABLISHED FOR MB1 — ADDED MODEL — one double-excitation channel

For C.1, retain only a bare two-cell reference $|0\rangle = |o,o\rangle$ (both particles in the reference orbital o) and one double excitation $|d\rangle$. A real cluster has many excited states; one denominator cannot replace that spectrum without a separate approximation.

$$H_X = \begin{pmatrix} 0 & v \\ v & \Delta \end{pmatrix}, \qquad v, \Delta > 0, \qquad \text{basis } (|0\rangle, |d\rangle).$$

C.1 Find the lower eigenvalue and a normalized lower eigenstate. Give their leading forms for $v/\Delta \ll 1$. Starting from the bare state $|0\rangle$, find the exact double-excitation probability $P_d(t)$ and its short-time form. Does this probability define an irreversible decay rate? **1.2 pt**

NOT ESTABLISHED FOR MB1 — ADDED MODEL — an overlap diagnostic

The following three-spin operators illustrate overlapping corrections on neighbouring clusters. They are illustrative and not fitted to any calculation:

$$h_{23} = \gamma\,\sigma_2^z\sigma_3^z, \qquad S_{12} = -i\epsilon\,\sigma_1^x\sigma_2^x, \quad \gamma, \epsilon \in \mathbb{R}.$$

C.2 Calculate $[h_{23}, S_{12}]$. Explain why a smaller residual on each isolated pair of cells does not, by itself, certify a small residual after the corrections are assembled along a chain. **0.6 pt**

Source / scope: Use $[\sigma_z, \sigma_x] = 2i\sigma_y$. Operators on different spins commute.

C.3. What survives the moving frame?

AVAILABLE — GIVEN TOOL — exact frame and source structure

A number-conserving one-particle frame lifts to the complete many-body Fock space (T7). It does not delete modes or replace interactions by a mean field. A two-body interaction can still produce single and double excitations of a bare Slater reference.

Let $W(t)$ be such a differentiable many-body frame. Set

$$\mathcal{K}_W = W^\dagger H_{\mathrm{MFP}} W - iW^\dagger \dot{W}.$$

For a differentiable anti-Hermitian many-body $S(t)$, let

$$\mathcal{U} = We^S, \qquad |\zeta\rangle = \mathcal{U}^\dagger |\Psi\rangle.$$

Use the identity

$$e^{-S}\partial_t e^S = \int_0^1 e^{-uS}\dot{S}e^{uS}\,\mathrm{d}u.$$

C.3 *Modelling bridge* **1.2 pt**

(a) Starting with $i\dot{\Psi} = H_{\mathrm{MFP}}\Psi$, derive the exact generator $\mathcal{K}_{\mathrm{corr}}$ of $|\zeta\rangle$ in terms of $\mathcal{K}_W, S, \dot{S}$.
(b) Identify the first-order terms if both S and its time derivative belong to a controlled perturbative expansion. Why may $\dot{S}$ not simply be removed?
(c) If $\mathcal{K}_{\mathrm{corr}} = \mathcal{K}_{\mathrm{ret}} + \mathcal{R}$, where does an approximation first occur? Write the transformed physical initial state. Explain why dropping $\mathcal{R}$ needs a bound.
(d) Does the one-particle frame turn H_{int} into a one-particle operator? Does writing e^S specify its coefficients or prove that it is local?

Hint. For (b), expand $e^{-S}\mathcal{K}_W e^S = \mathcal{K}_W + [\mathcal{K}_W, S] + \frac{1}{2}[[\mathcal{K}_W, S], S] + \cdots$.

Connection to the next step: C.4 tests whether the kept part actually preserves the desired reference.

C.4. Does “kept” mean “harmless”?

AVAILABLE — GIVEN TOOL — the kept generator

This question asks what the kept part $\mathcal{K}_{\mathrm{ret}}$ of the generator does to a chosen reference state.

For a fixed normalized direction $|A_0\rangle$ in the chosen frame, let $\Pi_A = |A_0\rangle\langle A_0|$. Assume $\mathcal{K}_{\mathrm{corr}} = \mathcal{K}_{\mathrm{ret}} + \mathcal{R}$ as in C.3.

C.4 *Modelling bridge* **0.9 pt**

(a) Give the necessary and sufficient instantaneous condition for $\mathcal{K}_{\mathrm{ret}}(t)$ to preserve $|A_0\rangle$ up to a phase throughout the interval.
(b) Return to the teaching spin chain of A.2, now with constant $g \neq 0$. Take $\mathcal{K}_{\mathrm{ret}} = H_{\mathrm{d}}$ and $\mathcal{R} = 0$. From the all-A state, compute $(H_{\mathrm{d}} - E_A)|A_0\rangle$, its squared norm, and the leading short-time term of $\langle q_j(t)\rangle$, where $q_j = (1 + \sigma_j^z)/2$.
(c) Has the retained model an approximation error in this example? Does zero omitted remainder prove that the reference direction is preserved? Does the short-time spin result give an MFP transport lifetime?

NOT ESTABLISHED FOR MB1 — ADDED MODEL — reuse A.2, do not infer an MB1 mapping

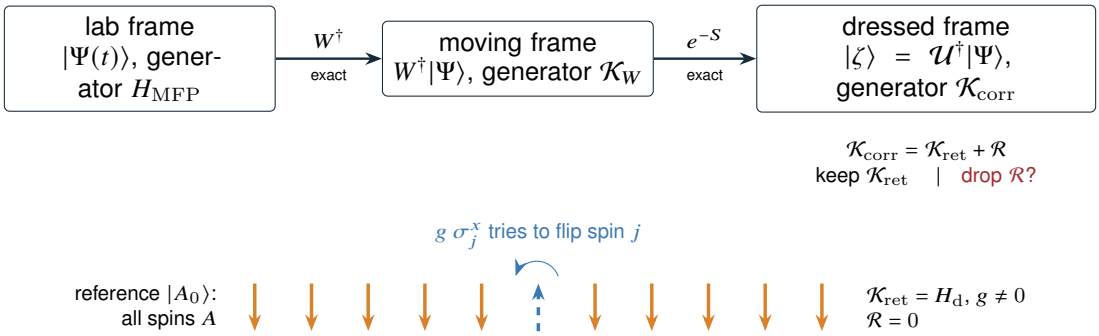
The spin chain in (b) is still the teaching analogy, not the full MFP Hamiltonian. Use a ring of N spins. Set $E_A = \langle A_0|H_\mathrm{d}|A_0\rangle$. Let $|j\rangle$ denote the orthonormal configuration with only spin j flipped. You may use $e^{-iHt} = \mathbb{1} - iHt + O(t^2)$.

A candidate reference evolution

A candidate may have the form

$$|\Psi_\mathrm{ref}(t)\rangle = W(t)e^{S(t)}|\Xi(t)\rangle, \qquad i\dot{\Xi} = \mathcal{K}_\mathrm{ret}(t)\Xi.$$

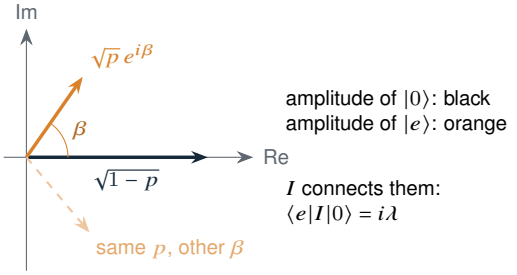
PICTURE 10 — C.3 and C.4 — changes of frame and what is kept

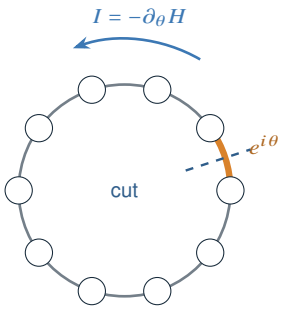


Top (C.3). Two exact changes of frame: first follow the geometry with W, then dress with correlations through e^S. C.3 asks for the exact generator in the dressed frame, and where the first approximation enters — the only candidate is dropping the part $\mathcal{R}$. **Bottom (C.4).** Suppose nothing is dropped ($\mathcal{R} = 0$). C.4 asks whether the kept generator can still move the reference state $|A_0\rangle$. In the spin example the transverse field g tries to flip each spin; part (b) asks how fast the flip probability of one spin starts to grow.

Part D. The physical current (2.0 points)

PICTURE 11 — D.1 and D.2 — current needs a phase; current through a twisted bond





Left (D.1). The probability p fixes the lengths of the two amplitudes; the phase β fixes their relative angle. D.1 asks how $\langle I\rangle$ depends on β and, at small p, on p, and whether p alone can fix the current. **Right (D.2).** On a ring, put a phase $e^{i\theta}$ on one bond (a "twist"). Then $I = -\partial_\theta H$ is the current through that bond. D.2 asks you to rewrite this same physical current after a change of frame $\mathcal{U}$, using $\mathcal{K}$ and $\mathcal{A}_\theta$, then to integrate it over time. The endpoint term $\langle \mathcal{A}_\theta\rangle$ is the piece one is tempted to drop.

NOT ESTABLISHED FOR MB1 — ADDED MODEL — a current-connected pair of states

For D.1, $|0\rangle$ and $|e\rangle$ are any orthogonal states connected by the stated current. Do not identify $|e\rangle$ with C.1's pure double excitation without checking the current matrix element. A one-body hopping current does not directly connect Slater determinants differing by two occupied orbitals.

$$I = \lambda \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad |\psi\rangle = \sqrt{1-p}|0\rangle + e^{i\beta}\sqrt{p}|e\rangle, \quad \lambda \in \mathbb{R}, \quad 0 \le p \le 1.$$

D.1 Find $\langle I\rangle$ and compare $\beta = 0, \pi/2, -\pi/2$. At small p, is the leading magnitude proportional to p or to $\sqrt{p}$? Can the probability p alone determine the current? **0.6 pt**

AVAILABLE — GIVEN TOOL — exact change of frame

The current must be transformed with the same unitary as the Hamiltonian. Work on one smooth gauge patch and use the actual endpoint frame if a patch boundary is crossed. No approximation to the many-body state is made in D.2.
Introduce a boundary twist θ: multiply the hopping on one chosen bond, from j to $j+1$, by $e^{i\theta}$ (and its conjugate by $e^{-i\theta}$). Then $I = -\partial_\theta H$ at $\theta = 0$ is the particle current through that bond. For a differentiable unitary $\mathcal{U}(\theta, t)$, define

$$\mathcal{A}_\mu = i\mathcal{U}^\dagger \partial_\mu \mathcal{U}, \qquad \mathcal{K} = \mathcal{U}^\dagger H \mathcal{U} - \mathcal{A}_t, \qquad \mu \in \{\theta, t\}.$$

You may use

$$\partial_\theta \mathcal{A}_t - \partial_t \mathcal{A}_\theta = i[\mathcal{A}_\theta, \mathcal{A}_t],$$

and the rule T3 for time-dependent operators.

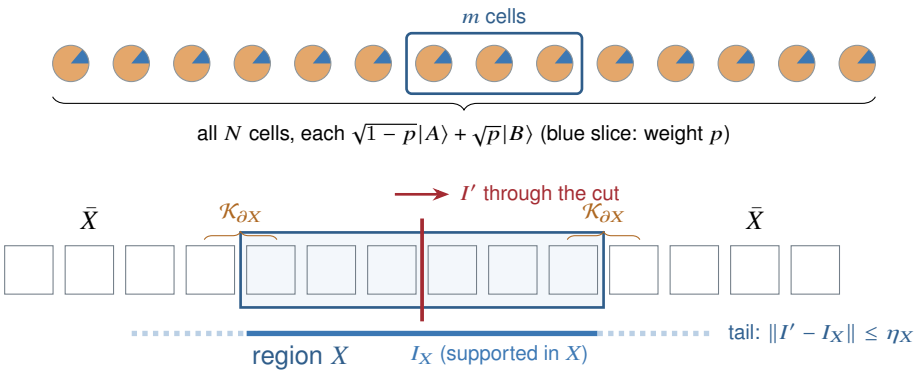
D.2 Prove that $\mathcal{U}^\dagger I \mathcal{U} = -\partial_\theta \mathcal{K} - \partial_t \mathcal{A}_\theta - i[\mathcal{K}, \mathcal{A}_\theta]$. For a state satisfying $i\partial_t|\Xi\rangle = \mathcal{K}|\Xi\rangle$, derive the integrated current from t_1 to t_2. Explain why a closed projector path does not justify deleting the endpoint term. **1.4 pt**

NOT ESTABLISHED FOR MB1 — OPEN — a current identity is not a current prediction

The identity is exact. The full MFP problem still requires the driven correlated state, its physical current, and any endpoint or boundary contribution.

Part E. Local accuracy in a large system (2.4 points)

PICTURE 12 — E.1–E.3 — global overlap versus a local region



Top (E.1). Every cell is slightly imperfect. E.1 compares the overlap with the ideal all-A state for the whole chain and for a fixed window of m cells, and asks what this says about global fidelity as a test of local transport. **Bottom (E.2, E.3).** The current is measured at the red cut. E.2 asks you to replace I' by an operator I_X supported in a region X around the cut, and to bound the current error by the distance between the two states reduced to X, plus the small tail η_X. E.3 asks which equation the reduced state σ'_X obeys; the terms $\mathcal{K}_{\partial X}$ that cross the edge of X are the reason the outside matters.

NOT ESTABLISHED FOR MB1 — ADDED STATE — a diagnostic product family

For E.1 only, prescribe the state below. It is not asserted to be a solution of the interacting pump. The scalar p is independent of N. Use the squared-fidelity convention $\mathcal{F}(\rho, |a\rangle) = \langle a|\rho|a\rangle$.

$$|\psi_N\rangle = \bigotimes_{j=1}^{N} \left[\sqrt{1-p}|A\rangle_j + \sqrt{p}|B\rangle_j\right], \qquad 0 < p < 1, \quad \langle A|B\rangle = 0.$$

E.1 Find its fidelity with $|A\rangle^{\otimes N}$. Then find the fidelity of an m-cell subsystem with $|A\rangle^{\otimes m}$, with m fixed as N increases. What does this comparison imply about using global fidelity to diagnose local transport? **0.5 pt**

AVAILABLE — GIVEN TOOL — local current-error structure

In E.2, use a local observable and reduced states rather than global many-body overlap. The inequality can be derived without a small-error or weak-interaction assumption.

Let ρ' and σ' be normalized density matrices in the same moving frame: ρ' is the exact transformed state and σ' the reference. Let I_X be supported near the measured cut, in a region X, and suppose $\|I' - I_X\| \le \eta_X$. Reduced density matrices on X are denoted by ρ'_X and σ'_X (T6). Set $\Delta I = \mathrm{Tr}[(\rho' - \sigma')I']$.

E.2 Prove $|\Delta I| \le \|I_X\| \left\|\rho'_X - \sigma'_X\right\|_1 + 2\eta_X$. If the reference is a product state in this frame, define $q_x = 1 - |A_{0,x}\rangle\langle A_{0,x}|$ and $D_X = \sum_{x\in X}\langle q_x\rangle_{\rho'}$. Deduce $|\Delta I| \le 2\,\|I_X\|\min(1, \sqrt{D_X}) + 2\eta_X$. State which form remains valid for a general correlated reference. **1.0 pt**

Useful facts

$$|\operatorname{Tr}(AB)| \le \|A\|_1 \|B\|, \qquad \|\rho - \sigma\|_1 \le 2,$$
$$\|\rho - |a\rangle\langle a|\|_1 \le 2\sqrt{1 - \langle a|\rho|a\rangle},$$
$$1 - \prod_{x\in X}(1 - q_x) \le \sum_{x\in X} q_x.$$

Source / scope: Here $\|\cdot\|$ is the operator norm and $\|\cdot\|_1$ is the trace norm. The local q_x are reference-deviation operators, not automatically physical domain counters.

E.3. From a kept Hamiltonian to a local state

AVAILABLE — GIVEN TOOL — same physical current, local states

The current-error inequality of E.2 uses two local reduced states in the same frame. It does not supply their dynamics. This question makes that missing input explicit, without introducing a heat bath.

Let the exact transformed state be ρ' and the reference state be σ'. Suppose

$$i\dot{\rho}' = [\mathcal{K}_{\mathrm{corr}}, \rho'],$$
$$i\dot{\sigma}' = [\mathcal{K}_{\mathrm{ret}}, \sigma'],$$
$$\mathcal{K}_{\mathrm{corr}} = \mathcal{K}_{\mathrm{ret}} + \mathcal{R}.$$

Both states are normalized. Their initial values must be stated; they need not coincide. For a region X near a cut, write

$$\mathcal{K}_{\mathrm{ret}} = \mathcal{K}_X + \mathcal{K}_{\bar{X}} + \mathcal{K}_{\partial X}, \qquad \sigma'_X = \operatorname{Tr}_{\bar{X}} \sigma'.$$

The first two terms act only inside and outside X. The third contains interactions that connect both regions.

E.3 *Modelling bridge* **0.9 pt**

(a) Derive the equation for σ'_X. Show why the contribution from $\mathcal{K}_{\bar{X}}$ vanishes under the partial trace.
(b) Is $\sigma'_X(0)$ and $\mathcal{K}_X(t)$ enough to solve that equation? Identify the additional object required by the boundary term. Does the composite-band gap supply it?
(c) Let $\delta\rho = \rho' - \sigma'$. Derive its exact equation and identify the initial-mismatch and omitted-dynamics inputs. Explain why rewriting this equation is not yet a bound on $\|\delta\rho_X(t)\|_1$.

Useful fact

For an operator $B_{\bar{X}}$ supported outside X, the partial trace is cyclic with respect to that operator:

$$\operatorname{Tr}_{\bar{X}}(B_{\bar{X}}A) = \operatorname{Tr}_{\bar{X}}(AB_{\bar{X}}).$$

Connection to the next step: D.2 supplies the correct current operator. E.2 converts a local-state error into a current error. E.3 identifies the missing dynamics between these two tools.

Part F. A conditional operating-time certificate (1.5 points)

NOT ESTABLISHED FOR MB1 — GIVEN BOUNDS — the missing long-time inputs

Let $\alpha, I_0 > 0$ and $\eta, \delta_0 \geq 0$. Assume that, for every $0 \leq s \leq t_{\max}$,

$$D_X(s) \leq \alpha^2 s^2, \qquad \|I_X(s)\| \leq I_0, \qquad \eta_X(s) \leq \eta.$$

Also assume that every relevant two-period reference window satisfies $|\overline{Q}_{\mathrm{ref}}(t) - Q_*| \leq \delta_0$. These remain supplied assumptions for the full correlated MB1 frame. In particular, the bound at $s = 0$ requires $D_X(0) = 0$. Initial mismatch must be controlled separately; do not silently replace the physical initial state by its projection. Here α, I_0, and η have units of inverse time; δ_0 and charge are dimensionless.

Define

$$\overline{Q}(t) = \frac{1}{2}\int_{t-2T}^{t} \langle I(s)\rangle \, \mathrm{d}s, \qquad 2T \leq t \leq t_{\max},$$

and $\overline{Q}_{\mathrm{ref}}(t)$ in the same way with the reference state. The factor is $1/2$, not $1/(2T)$: this is charge per drive period averaged over two periods, not mean current.

F.1 Use E.2 to bound $|\overline{Q}(t) - Q_*|$. For $\alpha I_0 > 0$, find a sufficient upper endpoint t_{cert} for which this bound is at most a tolerance δ. State the positivity and time-domain conditions. Evaluate the result with the illustrative inputs below. Does the next period necessarily fail? **0.7 pt**

NOT ESTABLISHED FOR MB1 — ILLUSTRATIVE NUMBERS — not MB1 measurements

Use the time unit $\tau_0 = T$, so $T/\tau_0 = 1$. Let

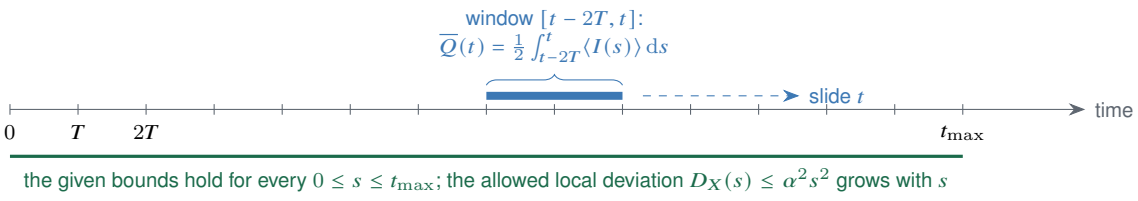
$$I_0\tau_0 = 1, \quad \alpha\tau_0 = 10^{-5}, \quad \eta\tau_0 = 10^{-5}, \quad \delta_0 = 0.001, \quad \delta = 0.005.$$

Assume $t_{\max} = 201\tau_0$. These numbers are chosen for arithmetic practice. They must not be quoted as an MFP lifetime result.

Connection to C.4

The product-reference form of E.2 compares ρ' with the *fixed* direction $|A_{0,X}\rangle$. It is therefore the right tool only while the reference state itself stays on that direction — exactly the property tested in C.4(a). C.4(b) showed that kept dynamics can move the reference even when $\mathcal{R} = 0$. In that case the premise $D_X(s) \leq \alpha^2 s^2$ for a fixed product reference is not the right input; one must use the general form of E.2 with an evolving reference, as in F.3.

PICTURE 13 — F.1 — a sliding two-period window



Every window of length $2T$ gives one averaged charge $\overline{Q}(t)$. The given bounds allow the local deviation from the reference to grow with time. **F.1 asks:** up to which window end t can you still guarantee $|\overline{Q}(t) - Q_*| \leq \delta$, and what can be said about the windows after that time?

F.3. Prediction, error, and first failure

AVAILABLE — GIVEN TOOL — the observable inequality

For the same frame and the same physical current, E.2 gives

$$|\Delta I(s)| \le \|I_X(s)\| \left\|\rho'_X(s) - \sigma'_X(s)\right\|_1 + 2\eta_X(s).$$

A reference charge must be calculated from its state. It is not assigned the target Q_* solely from a geometric winding.

Define $\widehat{Q}(t) = \frac{1}{2}\int_{t-2T}^{t} \mathrm{Tr}[\sigma'(s)I'(s)]\,\mathrm{d}s$. For this exercise, suppose computable functions satisfy

$$\left\|\rho'_X(s) - \sigma'_X(s)\right\|_1 \le e_X(s), \qquad \|I_X(s)\| \le b_X(s), \qquad \eta_X(s) \le \eta(s).$$

F.3 *Modelling bridge* **0.8 pt**

(a) Integrate the current inequality to obtain $|\overline{Q}(t) - \widehat{Q}(t)| \le E(t)$. State where initial mismatch must enter this construction.
(b) Derive sufficient tests to certify that a given window is inside or outside the tolerance $|\overline{Q} - Q_*| \le \delta$. What is the output when neither test holds?
(c) Apply the tests to the three supplied sample windows. Is the first failed sampled window determined? Is the first failure over continuously varying window starts determined?

NOT ESTABLISHED FOR MB1 — GIVEN FUNCTIONS AND NUMBERS — teaching inputs only

The functions above are not established MB1 bounds. For the arithmetic test take $Q_* = -0.5$ and $\delta = 0.005$:

Ordered sample window	$\widehat{Q}$	E
1	-0.4970	0.0010
2	-0.4940	0.0015
3	-0.4940	0.0005

These are invented classification examples, not new trajectories. Use one-window exceedance; no persistence rule is imposed in this small exercise.

PICTURE 14 — F.3 — a prediction with an error bar



A computed prediction $\widehat{Q}$ for one window, with a proven error E: the true $\overline{Q}$ lies on the blue bar. **F.3 asks:** for which positions of the bar relative to the grey band can you certify “inside”, certify “outside”, or neither? Then apply your rule to three given windows.

Connection to the next step: F.1 certifies an interval once special growth bounds are supplied. F.3 permits a reference that itself changes or fails. Both require model-derived inputs before either gives an MB1 prediction.

Part X. Synthesis: what would a prediction need? (0.5 points)

AVAILABLE — SUMMARY — the tools now on the table

The sources supply the candidate flavor mechanism [1] and projector kinematics [2]. This booklet adds exact frame, current and local-error identities. A full correlated pump is still missing.

NOT ESTABLISHED FOR MB1 — OPEN — the research task

Starting from $H_{\mathrm{MFP}}(t)$ and a specified physical initial state, construct a correlated reference evolution and a computable error estimate over complete drive cycles. The result must remain useful as the system size increases. Do not import F.1's long-time assumptions as facts.

Seek a prediction $\widehat{Q}(t)$ and an error $E(t)$ such that

$$|\overline{Q}(t) - \widehat{Q}(t)| \le E(t),$$

and the two tests

$|\widehat{Q} - Q_*| + E \le 0.005$ certifies that the window passes,

$|\widehat{Q} - Q_*| - E > 0.005$ certifies that the window fails.

X.1 List the inputs that must be derived from H_{MFP} and a specified physical initial state before either test can be applied to the original pump. Why does certifying individual windows not, by itself, locate the first failure? **0.5 pt**

What a correct answer does not prove

A solved toy model does not establish its equivalence to MFP. A checked matrix identity does not establish a small full-chain remainder. A certified lower bound is not an exact lifetime. These distinctions are part of the problem.

Sources

[1] M. Jürgensen, J. Steiner, G. Refael, and M. C. Rechtsman, *Multi-band fractional Thouless pumps*, arXiv:2504.09338v2 (2025). The original model and the flavor-following mechanism ("MFP"). `arXiv:2504.09338`.

[2] S.-M. Huang, *Quantum-State Projectors on Grassmannian: Geometry, Holonomy, and Topology*, arXiv:2608.06777v1 (2026). Projectors and their generators ("H2"). `arXiv:2608.06777`.

[3] APhO 2025 theoretical question booklet, used as a layout reference only. No exam text is reproduced.

This public example uses only these sources and teaching models.

Optional background reading (public; not needed for any question)

- Thouless pumps: R. Citro and M. Aidelsburger, *Thouless pumping and topology*, Nat. Rev. Phys. **5**, 87 (2023), `arXiv:2210.02050`.
- Berry phase and adiabatic transport: D. Xiao, M.-C. Chang, and Q. Niu, Rev. Mod. Phys. **82**, 1959 (2010), `arXiv:0907.2021`.
- Wannier functions and Wilson loops: N. Marzari *et al.*, Rev. Mod. Phys. **84**, 1419 (2012), `arXiv:1112.5411`.
- Exponential localization: C. Brouder *et al.*, PRL **98**, 046402 (2007), `arXiv:cond-mat/0606726`.
- Dressing transformations e^{S}: S. Bravyi, D. P. DiVincenzo, and D. Loss, Ann. Phys. **326**, 2793 (2011), `arXiv:1105.0675`.
- Domain formation in spin chains: G. Lagnese *et al.*, Phys. Rev. B **104**, L201106 (2021), `arXiv:2107.10176`.
- Trace distance and fidelity: C. A. Fuchs and J. van de Graaf, `arXiv:quant-ph/9712042`.
- Position operator on a ring (used for Pictures 2): R. Resta, Phys. Rev. Lett. **80**, 1800 (1998).

NOT ESTABLISHED FOR MB1 — The main scientific gap

Neither source supplies a controlled full-cycle correlated reference, its physical current, or a quantitative large-system lifetime. Red frames mark those missing links.

The solution booklet contains worked answers and marking notes.

End of question booklet